Network commands

Network commands are very useful for troubleshooting issues on the network and to check the particular port connecting to the client.

ping	Send ICMP packets
traceroute	Print route packets to a network
netstat	print network connection, routing table, interface stats
ftp/lftp	Internet file transfer program
wget	Non Interactive network downloader
ssh	OpenSSH SSH Client (remote login program)
scp	secure copy
sftp	Secure File transfer program

Grep commands

Grep commands are useful to find the errors and debug the logs in the system. It is one of the powerful tools that shell has.

grep -h '.zip' file.list	. is any character
grep -h '^zip' file.list	starts with zip
grep -h 'zip$' file.list	ends with zip
grep -h '^zip$' file.list	containing only zip
grep -h '[^bz]zip' file.list	not containing b and z
grep -h '^[A-Za-z0-9]' file.list	file containing any valid names

Quantifiers

Here are some examples of quantifiers:

?	match element zero or one time
*	match an element zero or more times
+	Match an element one or more times
{}	match an element specfic number of times

Text processing

Text processing is another important task in the current IT world. Programmers and administrators can use the commands to dice, cut and process texts.

cat -A $FILE	To find any CTRL character introduced
sort file1.txt file2.txt file3.txt > final_sorted_list.txt	sort all files once
ls - l | sort -nr -k 5	key field 5th column
sort --key=1,1 --key=2n distor.txt	key field 1,1 sort and second column sort by numeric
sort foo.txt | uniq -c	to find repetition
cut -f 3 distro.txt	cut column 3
cut -c 7-10	cut character 7 - 10
cut -d ':' -f 1 /etc/password	delimiter :
sort -k 3.7nbr -k 3.1nbr -k 3.4nbr distro.txt	3 rd field 7 the character, 3rd field 1 character
paste file1.txt file2.txt > newfile.txt	merge two files
join file1.txt file2.txt	join on common two fields

Hacks and tips

In Linux, we can go back to our history of commands by either using simple commands or control options.

clear	clears the screen
history	stores the history
script filename	capture all command execution in a file

```
Tips:

History    : CTRL + {R, P }
!!number : command history number
!!          : last command
!?string    : history containing last string
!string      : history containing last string
export HISTCONTROL=ignoredups
export HISTSIZE=10000
```

As you get familiar with the Linux commands, you will be able to write wrapper scripts. All manual tasks like taking regular backups, cleaning up files, monitoring the system usage, etc, can be automated using scripts. This article will help you to start scripting, before you move to learning advanced concepts. END

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